

Python Practice Memo

1. Hello World!

Print "Hello World"

```
print("Hello World")
```

2. Hello, User! V1

Create a variable "name" and assign your name to the variable. Then print the variable.

```
name = "John Doe"  
print(name)
```

3. Rectangle

Print a rectangle using 'O.' (Length: 5; Height: 4)

```
print("00000")  
print("00000")  
print("00000")  
print("00000")
```

4. Right-triangle

Print a right-triangle using '*'. The height is 5, and the base is 5.

```
print("*")  
print("**")  
print("***")  
print("****")  
print("*****")
```

5. Square

Print a square using the char that a user input. (Length: 4; Height: 4)

```
print(char + char + char + char)  
print(char + char + char + char)  
print(char + char + char + char)  
print(char + char + char + char)
```

OR

```
print(char * 4)  
print(char * 4)  
print(char * 4)  
print(char * 4)
```

6. Hello, User! V2

Write a Python program that asks the user for their name and then prints a greeting message including their name.

```
name = input("What is your name? ")  
print("Hello, " + name + "!")
```

7. Addition Calculator

Create a simple calculator that takes two numbers as input from the user and prints the sum of those numbers. Assume the user will only work with Integers.

```
number1 = int(input("Enter the first number: "))  
number2 = int(input("Enter the second number: "))  
  
sum = number1 + number2  
  
print("The sum of the numbers is: ", sum)
```

8. Area of a Rectangle

Write a program that asks the user for the width and height of a rectangle and then prints the area of the rectangle.

```
width = int(input("Enter the width of the rectangle: "))  
height = int(input("Enter the height of the rectangle: "))  
  
area = width * height  
  
print("The area of the rectangle is: ", area)
```

9. Temperature Converter

Create a program that converts temperatures from Fahrenheit to Celsius.

Formula: $(\text{Fahrenheit} - 32) \times \frac{5}{9}$

```
fahrenheit = float(input("Enter the temperature in Fahrenheit: "))  
celsius = (fahrenheit - 32) * 5/9  
  
print("The temperature in Celsius is: ", celsius)
```

10. Simple Interest Calculator

Write a program that calculates simple interest given the principal amount, the rate of interest, and the time (in years).

Use the formula: $\text{Simple Interest} = (\text{Principal} * \text{Rate} * \text{Time}) / 100$.

```
principal = float(input("Enter the principal amount: "))
rate = float(input("Enter the rate of interest: "))
time = float(input("Enter the time (in years): "))

simple_interest = (principal * rate * time) / 100

print("The simple interest is: ", simple_interest)
```